Lab 8: Combinations, neural networks and foundation models

MATH840 | Time Series | Weeks 9–10 — graded assignment 7 of 7

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The last lab — your choice of weapon

Your series from Week 4, one last time. You have an ETS model from Lab 5 and a seasonal ARIMA from Lab 7, and a published accuracy score for each. Those are **the models to beat**. The last two weeks of lectures gave you three ways to try — pick one, two, or all of them, and find out with your own numbers whether anything new was worth it.

Open the notebook

The template: Open in Colab

Click **Copy to Drive** before you start working, otherwise nothing you write is saved. A CPU runtime is enough for every track.

The practice sessions — case 5 in each is this lab on a public series:

- Week 9, Prophet and combinations: Open in Colab · Ukrainian version with results
- Week 10, foundation models: Open in Colab · Ukrainian version with results

The task

1. **Rebuild** your Lab 5 and Lab 7 models exactly as you submitted them, add MSTL, and run them on four rolling windows next to your benchmark. Everyone does this.
2. **Choose your track** — and write down what you expect *before* you run it:

Track	The idea	Lecture
A. Combine	an equal-weight average of your models	*Advanced Forecasting Methods*
B. Train	MLP, N-BEATS, LSTM, trained on your series	*Neural Networks for Forecasting*
C. Borrow	Chronos-2, TimesFM 2.5, TiRex, zero-shot	*Foundation Models for Time Series*

Track	The idea	Lecture
D. Your own way	all of the above, a mix of old and new, tuning, another model — anything you can defend	—

3. **One scoreboard** for everything you ran, on the same four windows.
4. **Submit** what the scoreboard supports, with an 80% interval checked on windows it did not see.
5. **Conclude** against your Lab 5 and Lab 7 results — and say what you would use at work.

One track done well is a full-marks lab. More tracks, done with care, count toward the bonus.

Not this time: plots of the series, decomposition, ACF. Every plot is about the models.

> ⓘ **Already submitted the combination lab?**
>
> The earlier version of Lab 8 is track A of this one. If you submitted it, it stands as it is — you may resubmit with the new template by the deadline to take another track; the later submission counts.

> ⓘ **This one counts**
>
> 8 points. **Deadline: Sunday 1 November 2026, 23:59**, on Moodle. The accuracy score is published the next morning.

📋 What you are marked on

Section	Criteria	Points
1	Your models, rebuilt — the ones to beat	1
2	Forecast accuracy against your benchmark	3
3	Justification block	3
4	Code quality and reproducibility	1
	Total	**8**

The notebook has one section per row. **Keep those headings.**

🔁 Section 1 — Your models, rebuilt (1 point)

- `MEMBERS` holds your **submitted** Lab 5 ETS and Lab 7 seasonal ARIMA — the specification, the damping, the log — and MSTL, with a log decision of its own.

- `PAST` holds the hidden-horizon MASE and skill you were sent after Labs 5 and 7.
- The validation-window MASE **must match your earlier notebooks.** If it does not, find out why and say so.
- Two or three sentences: do the numbers match, and how do the three models stand on the four windows?
- Did not submit Lab 5 or Lab 7? Use the automatic model and say so; it costs nothing.

🎯 Section 2 — Forecast accuracy (3 points)

What you may submit: **any row of your scoreboard** — one forecaster, or an average you built with `combine` — with an 80% interval, its own or conformal. Chosen on the rolling windows, never on anything else. The template's `final_forecast` produces it from the whole history for you.

Scored from your submission file on the `h` observations after your history that nobody has seen:

$$\text{skill} = \frac{\text{MASE of your forecast}}{\text{MASE of your benchmark}}$$

Skill	Points
below 1 — you beat your benchmark	3
1 to 1.1 — within 10%	2
above 1.1	1, and 3.3 must say why

> ⚠ **One hard rule: train on your own series only**
>
> The Track B pool cuts several series from the same source, disguised. **Another student's history can contain your hidden horizon.** A network trained across the other series would be trained on your answer. Not allowed, and not needed: the foundation models bring the "many series" experience with them.

> ⓘ **What the pool says**
>
> Measured on all 140 issued series, each forecast made from the full served history and scored on the hidden horizon. ETS and ARIMA are the automatic ones; the networks use the template's settings.

Forecast	Beats the benchmark	Within 10%	Worse	Median skill	Worst skill
`AutoETS` + `AutoARIMA` + Chronos-2	**99%**	1%	0%	**0.785**	1.08
`AutoETS` + `AutoARIMA` + `MSTL`	96%	3%	1%	0.798	1.18

Forecast	Beats the benchmark	Within 10%	Worse	Median skill	Worst skill
`AutoETS` + `AutoARIMA`	100%	—	—	0.805	0.93
select the best of `AutoETS`, `AutoARIMA`, `MSTL` on four windows	96%	2%	2%	0.811	1.23
`MSTL` alone	80%	6%	14%	0.812	2.46
Chronos-2	84%	9%	7%	0.819	2.05
TimesFM 2.5	76%	10%	14%	0.837	1.61
TiRex	77%	11%	11%	0.869	2.37
N-BEATS, trained on the one series	57%	12%	31%	0.939	3.02
MLP, trained on the one series	50%	14%	36%	0.997	3.48
LSTM, trained on the one series	46%	9%	45%	1.041	4.14

- **The pool was screened so that `AutoETS` and `AutoARIMA` each beat the benchmark** — so every row with them is flattered by construction. MSTL, the networks and the foundation models were not screened: their rows are what you would expect on fresh data.
- **A network trained on one series is close to a coin flip** against the benchmark: a few hundred observations are little to learn from.
- **A foundation model alone** beats the benchmark on 76–84% of the series and can lose badly on the rest. **Added to the old models** it gave the best median of all — the Week 9 lesson again.
- **Intervals of `AutoETS` + `AutoARIMA` + `MSTL`, on the hidden horizon:** the averaged bounds cover 78% of the points on average, the conformal recipe 79%; Winkler scores 3.75 and 3.85. Neither recipe wins in general — which is why you choose on your own windows.
- Your Lab 5 and Lab 7 models may not be the automatic ones. The table is a guide, not a promise.

✍ Section 3 — Justification block (3 points)

The part a human reads closely. Roughly a paragraph each, with numbers from your own cells.

Item	Points	What earns them
3.1	1	**Your track.** What you tried and what the windows show against your Section 1 models and the benchmark. *A:* members chosen before the table, windows won, error correlations, combine or select. *B:* the networks window by window, how many observations they learned from, what `input_size` means for your series, what a second seed changed, the cost in seconds. *C:* the three models window by window, their cost, and whether you can rule out that one saw your series' source. *D:* what question your extra work answered, and the answer. Several tracks — a short paragraph each.
3.2	1	**The interval.** The recipes for your choice on unseen windows: coverage, Winkler score, width. The one you submitted and why. How many errors the conformal radius rests on, and what that means for trusting it.
3.3	1	**What you submit, against the whole course.** Your Lab 5 and Lab 7 hidden-horizon skills next to the rolling evidence. What you submitted and why; the skill you expect; how it compares with what you expected in 2.0. Which approach you would use if you forecast this series at work every month, and what you would check.

Important

Marked on the reasoning, not on which track wins. "Nothing new beat my ETS, so I submitted it" earns full marks if the numbers show it and you explain why. So does "a foundation model won three windows of four, here is why I trust it — or don't".

Track C: what the model cards say they were trained on

You do not know your series' source, but you know the families: Track B is cut from public Australian and US economic statistics.

Model	Published training data
Chronos-2	Technical report, Table 6: 22 real datasets (M4 monthly, electricity and others) plus synthetic data
TimesFM 2.5	Model card: GIFT-Eval pretraining set (GIFT-Eval paper, Table 14: Monash datasets incl. M1, M3, tourism, Australian electricity, ...), Wikimedia pageviews, Google Trends, synthetic data
TiRex	Paper, Tables 5–6: subsets of the Chronos and GIFT-Eval training sets

None of them lists Australian retail turnover or US employment statistics. "Not listed" is not "not seen": a corpus can contain a series under another name. Say what you can and cannot rule out.

Where this goes wrong

- **Choosing on the validation window alone.** Four windows, and the scoreboard's mean — not window 1.
- **Choosing the members or the mix after reading the table.** Then it is a selection in disguise. Write the notes in 2.0 and 2.1 first.
- **Reading one seed as the network's score.** A small network on one short series can move a lot from seed to seed. Run your best one twice.
- **"New" read as "better".** On many Track B series ETS or ARIMA is still the one to beat.
- **Judging an interval on the windows it was built from.** `interval_check` leaves each window out.
- **Forecasting from `train`.** `final_forecast` calls every forecaster again on the whole of `y`.
- **An API key in the notebook.** TimeGPT is welcome in track D; the key comes from a Colab secret.

📖 FPP3 §13.4 Forecast combinations · §12.4 Neural network models · §5.5 Prediction intervals · Chronos-2 technical report · TimesFM 2.5 model card · TiRex paper

🖌 Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- The submission file is written **by the notebook**, not edited by hand.
- No leakage: everything is compared on the rolling windows; only the final forecast sees all of `y`; no network is trained on any series but yours.
- Every number in your text comes from a cell.

📤 Submission

Upload three files to the Lab 8 activity on Moodle **by Sunday 1 November 2026, 23:59**:

- `SURNAME_lab09_forecast.csv` — written by the last cell of the template. Exactly h rows and these columns:

Column	What
`student_id`	your identifier
`code`	your series code, the same as before
`ds`	the dates you are forecasting, in order
`yhat`	your point forecast
`yhat_lo_80`, `yhat_hi_80`	your 80% interval

- `SURNAME_lab09.pdf` — the report, with the justification block;
- `SURNAME_lab09.ipynb` — the source.

⚠ **A file that cannot be read scores 0 for accuracy**

> Exactly h rows, the dates asked for, no `NaN`, the interval around the point forecast. The template's check cell is the same one the scorer uses — if it passes, your file is fine.

⏱ How long this should take

About two hours for one track. Section 1 refits three models on four windows — a few minutes for a monthly ARIMA. Track B trains 12 networks — a few minutes on a CPU; track C needs seconds once the weights are downloaded. Most of the time should go into Section 3.